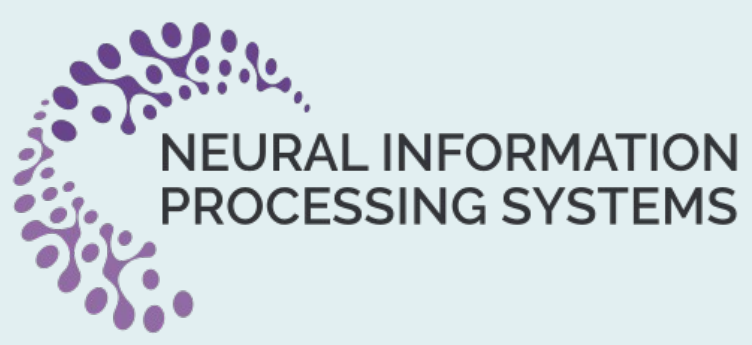


Neural Unbalanced Optimal Transport via Cycle-Consistent Semi-Couplings

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Problem

Predict how an **unpaired** cell population evolves under a drug when measurement is **destructive** — the same cell can never be profiled twice, so correspondences must be inferred from unpaired snapshots.

Classical optimal transport assumes **conserved mass** — it breaks exactly when cells proliferate or die and the population size shifts.

Motivation

- Drug responses are **heterogeneous**: within one sample some subpopulations proliferate while others die — aggregate models miss this.
- Prior neural OT either assumes **balance**, or (SOTA **ubOT GAN**) only captures the coarse growth trend, not per-cell reweighting.

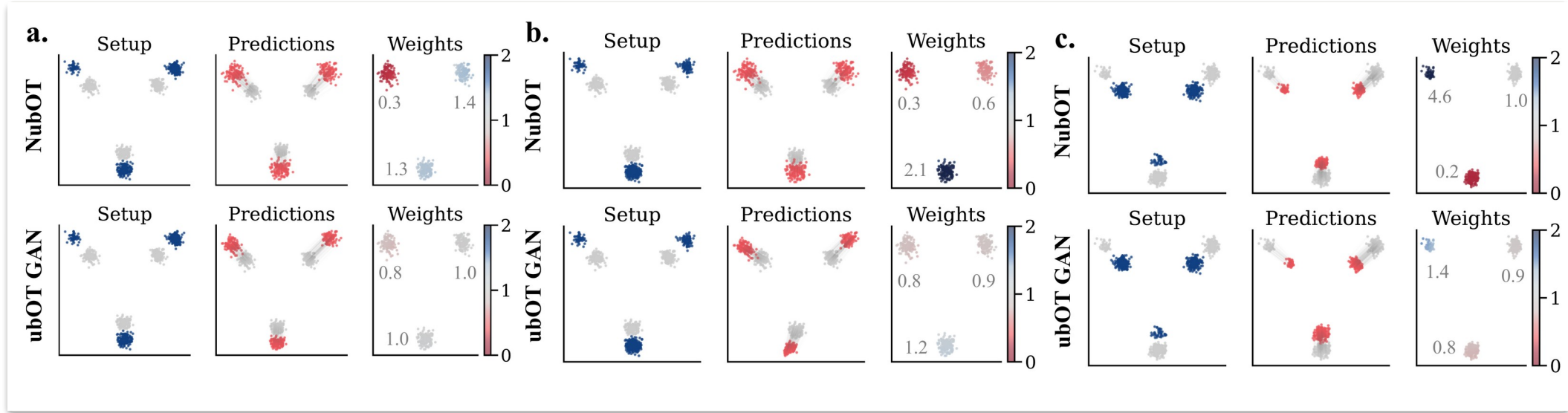


Fig 2 · Synthetic mapping. Across three imbalance scenarios (a–c), both methods map clusters to the right location, but only NubOT predicts the per-cluster weights (right panels) needed to match the target.

Method

- Reweight to balance.** Learned fields η, ζ scale source and target into equal-mass proxies $\tilde{\mu} = \eta \cdot \mu, \tilde{\nu} = \zeta \cdot \nu$.
- Transport & alternate.** Balanced OT via convex **ICNN** Monge maps $T = \nabla g, S = \nabla f$; alternate Sinkhorn reweighting with min–max potential updates; new cells map with predicted mass.

$$\tilde{\mu} = \eta \cdot \mu, \quad \tilde{\nu} = \zeta \cdot \nu, \quad T_{\#} \tilde{\mu} = \tilde{\nu}, \quad S_{\#} \tilde{\nu} = \tilde{\mu}$$

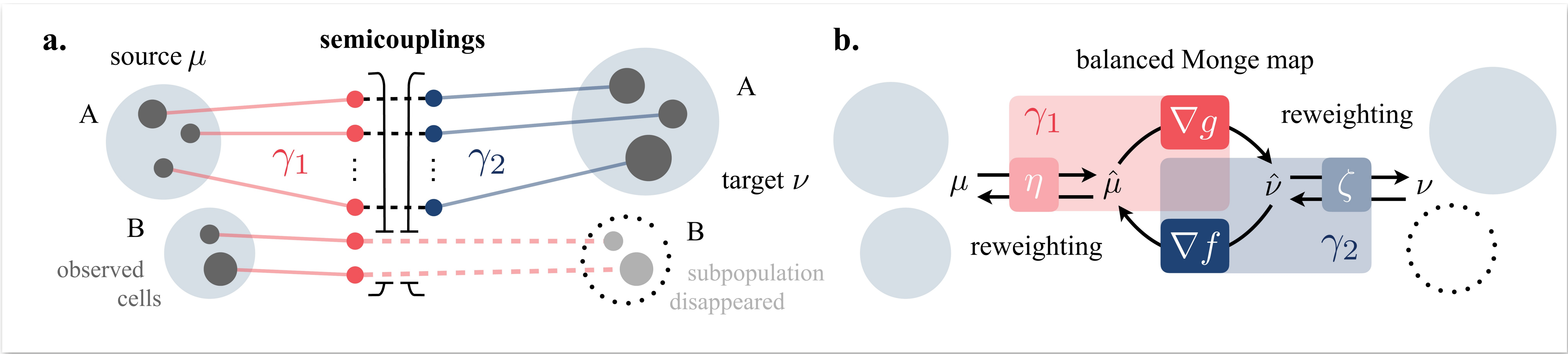


Fig 1 · Semi-couplings. (a) A pair (γ_1, γ_2) solves unbalanced OT — γ_1 where mass leaves μ , γ_2 where it arrives in ν ; a subpopulation can vanish. (b) NubOT builds them from reweighting η, ζ and the dual potentials f, g .

Key Results

METHOD	FEAT.	MASS
CellOT	✓	✗
ubOT GAN (SOTA)	✓	~
NubOT (Ours)	✓	✓

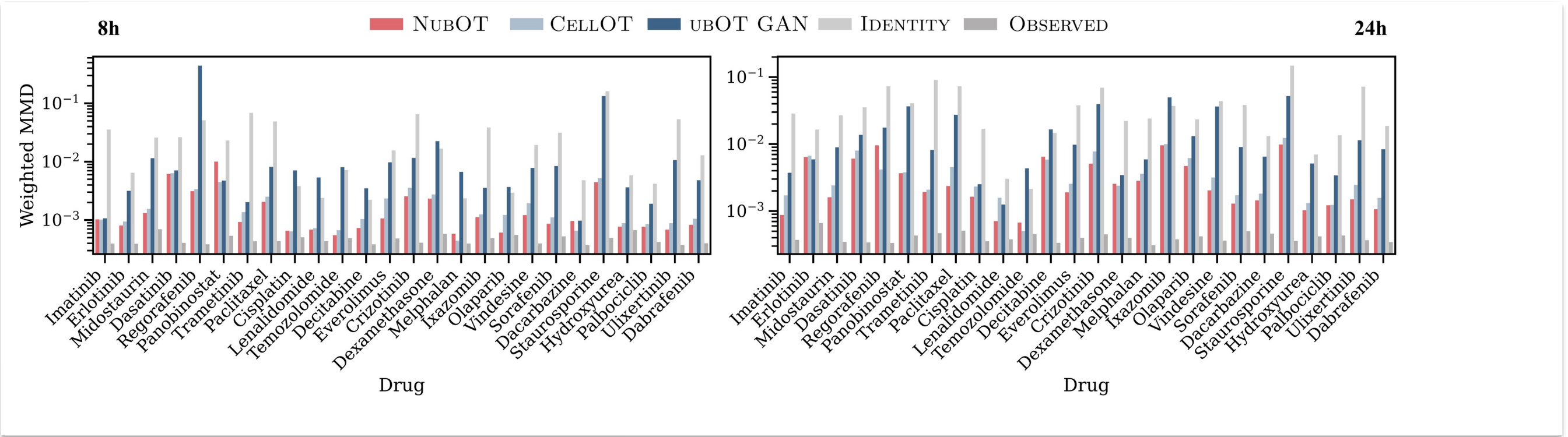


Fig 3 · Distributional fit. Weighted kernel MMD per drug at 8h & 24h — NubOT (red) lowest across almost all 25 drugs, beating every baseline

Biological Validation

- Predicted weights track observed cell-count fractions per subpopulation at 8h (**R = 0.95**).
- ≥ 1 where Ki67 (proliferation) is high, ≤ 1 where ClCasp3 (death) is high.

Recovers real proliferation and death signals, not superficial fit.

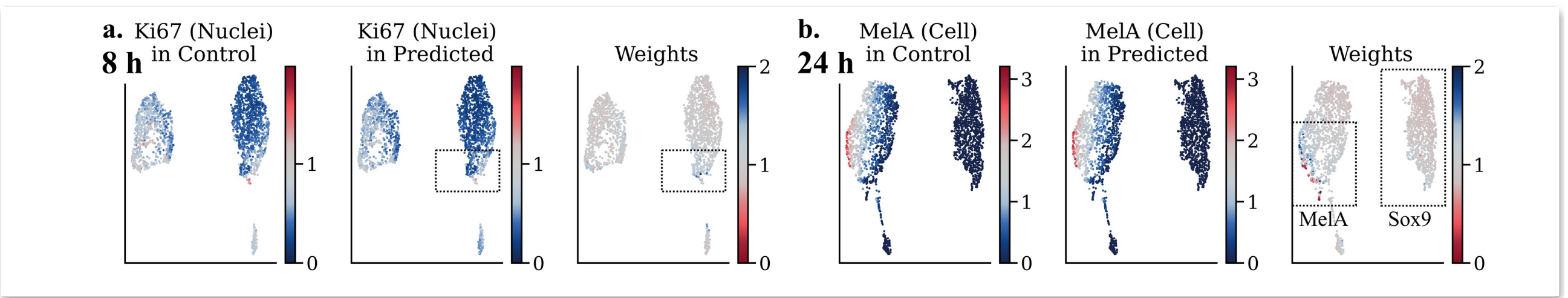


Fig 4 · Ulixertinib. UMAP by markers & predicted weights: weights > 1 for proliferating MeIA⁺ cells, death in Sox9⁺ cells by 24h

Dataset / Benchmark

Single-cell **melanoma drug-response** (4i imaging) + a synthetic imbalance benchmark.

25 drugs 0h · 8h · 24h MeIA⁺ Sox9⁺

Headline Numbers

R = 0.95
weight–count correlation, 8h
 $P = 2.2 \times 10^{-25}$

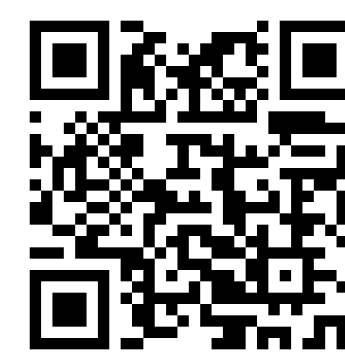
25 drug perturbations won
0.44 weight–count R at 24h
2 melanoma cell lines

Takeaway

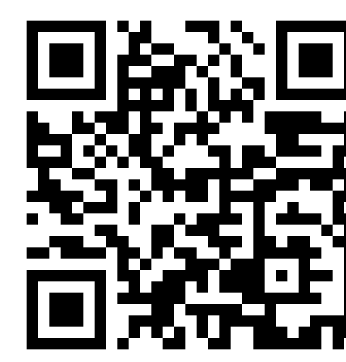
By recasting unbalanced OT through learnable **semi-couplings** and cycle-consistent training, NubOT jointly predicts **where cells move and how their mass grows or dies** — while staying biologically interpretable.

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Paper



Code



arxiv.org/abs/2209.15621 · github.com/FrederikeLuebeck/nubot